

Beyond Point Forecasts: Multi-Task Learning for Macroeconomic Tail Risk

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January 13, 2026

Assessing macro-financial tail risks still challenging

Standard macro-fin toolkit: GDP-at-risk (Adrian, Boyarchenko, and Giannone 2019); unemployment-at-risk (Kiley 2022); inflation-at-risk (López-Salido and Loria 2024)

1. How to fully capture non-linearity and possible skewness?

- ▶ Adrian, Boyarchenko, and Giannone (2019), Kiley (2022), and López-Salido and Loria (2024) use quantile regression (conditional linear)

2. How to allow for big data?

- ▶ Adrian, Boyarchenko, and Giannone (2019) assume only an index of financial conditions matter
- ▶ Adrian, Chen, et al. (2025) and Chi et al. (2025) use Quantile ML

3. How to ensure coherent joint movement of tails?

- ▶ Quantile VAR (Chavleishvili and Manganelli 2024)
- ▶ Modelling quantile spacing (Schmidt and Zhu 2016) prevents quantile crossing

Challenges can be solved with our new ML method

How do we assess macro-financial tail risks in this paper?

- ⇒ **Deep learning**: allows for high degree of time-varying complexity and high-dimensional data
- ⇒ **Multi-task**: use info learned by each task to learn other tasks (Caruana 1997)
- ⇒ **No-crossing architecture**: implement quantile spacings (Schmidt and Zhu 2016)

Contributions

1. Develop multi-task neural networks for joint quantile forecasting that respect monotonic quantile ordering
2. Forecast distributions of production/unemployment/inflation across horizons
3. Better *out-of-sample* accuracy than quantile ML and econometric models
4. Show joint quantile forecasts increase accuracy
5. Show quantile forecasts lead to good mean forecasts
6. *Will investigate tail risk drivers compared to the literature*

Closest paper to us: Adrian, Chen, et al. (2025)

What they do: forecast the tails ($\tau = 0.05$ and $\tau = 0.95$) of monthly industrial production growth, one-month ahead using Quantile Random Forest (QRF), Quantile Gradient Boosting (QGB), Quantile Partial Correlation Regression (QPCR), GARCH, Quantile Regressions (QR)

How we differ:

- ▶ Add deep learning
- ▶ Introduce multi-task learning
- ▶ Introduce no-quantile crossing
- ▶ Explore industrial production, inflation and unemployment
- ▶ We consider 5 quantiles and explore possible (shared) drivers
- ▶ We explore shorter and longer horizon (1 and 4 quarters ahead)

Data and Methodology

- ▶ 116 stationary predictors from FRED-MD over Jan 1961–Dec 2024.
- ▶ Response variables:
 - ▶ Inflation: $\pi_{t+h} = \ln(CPI_{t+h}) - \ln(CPI_t)$.
 - ▶ Industrial Production growth: $IP_{t+h} = \ln(IP_{t+h}) - \ln(IP_t)$.
 - ▶ Pct change in unemployment rate: $UR_{t+h} = \ln(UR_{t+h}) - \ln(UR_t)$.
- ▶ Annual expanding window training procedure beginning with Jan 1961–Dec 1997 to forecast Jan 1998–Dec 1998.

Data and Methodology

- Quantile forecasts are evaluated with the out-of-sample R^1 metric:

$$R_{oos}^1 = 1 - \frac{\sum_{t=1}^{T_{oos}} \rho_{\tau}(y_{t+h} - \hat{y}_{\tau,t+h})}{\sum_{t=1}^{T_{oos}} \rho_{\tau}(y_{t+h} - \bar{y}_{\tau})}$$

where $\bar{y}_{\tau}$ is the expanding historical quantile

- Mean forecasts are evaluated with the out-of-sample R^2 metric:

$$R_{oos}^2 = 1 - \frac{\sum_{t=1}^{T_{oos}} (y_{t+h} - \bar{Q}_{t+h})^2}{\sum_{t=1}^{T_{oos}} (y_{t+h} - \bar{y})^2}$$

where $\bar{y}$ is the expanding historical mean

and $\bar{Q}$ is the robust estimate of the mean taken from Liu (2023):

$$\bar{Q} = 0.15Q_{0.05} + 0.225Q_{0.25} + 0.25Q_{0.5} + 0.75Q_{0.75} + 0.15Q_{0.95}$$

Data and Methodology

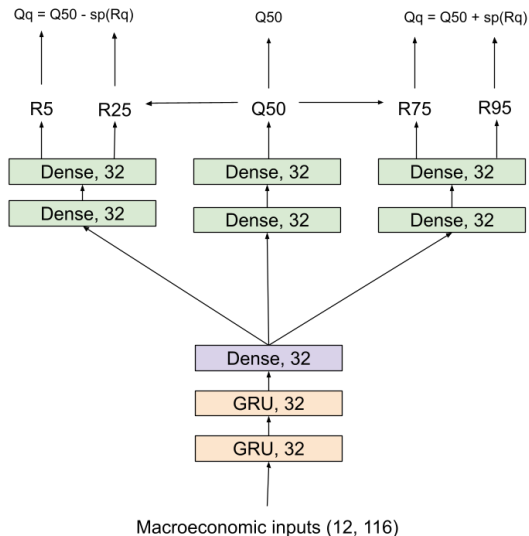


Figure 1: DMQv1 architecture (total trainable parameters 29,349)

Main Results – Inflation

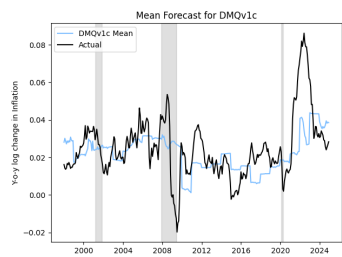
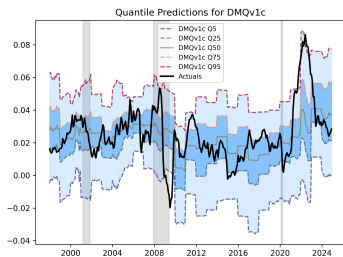
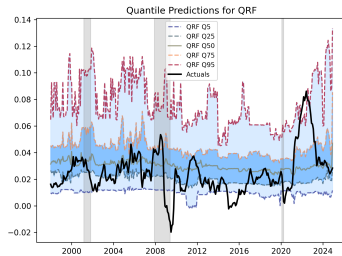
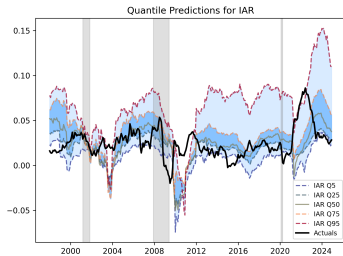
Table 1: Year-over-year Inflation 4 quarters ahead R_{oos}^1 (%) for Jan 1998–Dec 2024

	AR1	IAR	LR	LAS	QRF	QGB	DMQv0c	DMQv1c	DMQv2c
Mean	1.0	-25.5	-160.4	-56.3	13.7	2.1	20.9	27.6	15.2
Q5	1.4	-48.5	-466.0	-45.5	-1.4	-58.0	16.0	-12.6	-16.8
Q25	2.0	-19.1	-51.4	-8.1	13.6	-3.2	12.2	16.7	2.7
Q50	0.9	-20.5	-33.7	2.6	12.1	0.4	19.5	28.7	16.8
Q75	0.8	-17.7	-19.1	4.4	17.3	32.5	31.0	42.5	32.2
Q95	-0.3	-21.9	-231.7	-235.1	26.9	38.9	25.6	62.8	41.1

Table 2: R_{oos}^2 (%) for Jan 1998–Dec 2024

Model	AR1	IAR	LR	LAS	QRF	QGB	DMQv0c	DMQv1c	DMQv2c
R2	-7.3	-10.4	-680.5	-165.5	29.4	43.7	42.1	57.1	46.3

Main Results – Inflation



Main Results – Industrial Production

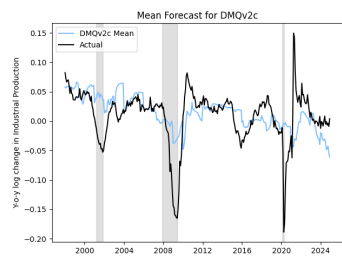
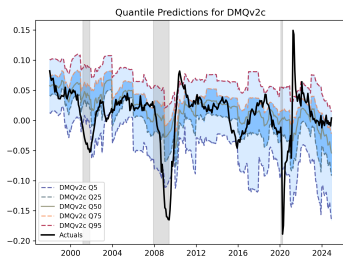
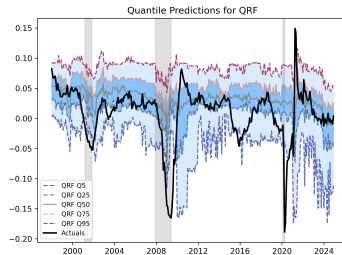
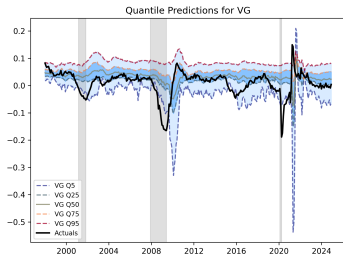
Table 3: Year-over-year IP growth 4 quarters ahead R_{oos}^1 (%) for Jan 1998–Dec 2024

	AR1	VG	LR	LAS	QRF	QGB	DMQv0c	DMQv1c	DMQv2c
Mean	0.6	-1.7	-126.6	-16.3	5.5	-7.5	9.6	4.6	13.4
Q5	1.7	-10.7	-116.3	-47.0	0.1	-76.8	2.3	-4.2	17.9
Q25	0.8	-4.7	-47.1	0.2	1.6	-5.2	5.8	2.1	10.1
Q50	0.6	-3.9	-72.3	-8.5	4.8	4.9	10.6	5.7	9.6
Q75	0.2	6.7	-43.9	-0.3	11.9	27.9	22.5	15.7	19.7
Q95	-0.1	4.2	-353.6	-26.0	9.3	11.7	6.8	3.9	9.5

Table 4: R_{oos}^2 (%) for Jan 1998–Dec 2024

Model	AR1	VG	LR	LAS	QRF	QGB	DMQv0c	DMQv1c	DMQv2c
R2	1.5	-11.5	-180.0	-15.7	3.9	6.2	14.1	14.6	23.4

Main Results – Industrial Production



Main Results – Unemployment

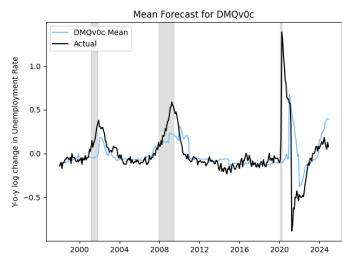
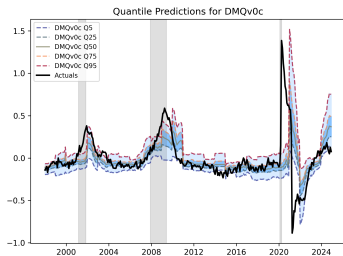
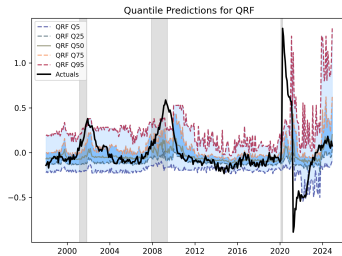
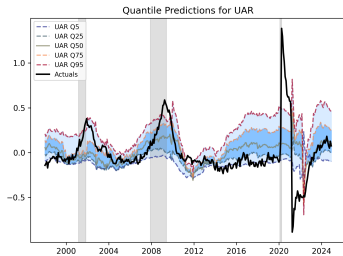
Table 5: Year-over-year Unemployment rate "growth" 4 quarters ahead R_{oos}^1 (%) for Jan 1998–Dec 2024

	AR1	UAR	LR	LAS	QRF	QGB	DMQv0c	DMQv1c	DMQv2c
Mean	0.8	-1.4	-80.9	-12.2	14.5	1.9	17.1	9.7	6.9
Q5	1.1	-19.3	-188.7	-7.3	20.4	-8.1	16.8	-9.2	-22.5
Q25	0.3	-12.3	-73.9	2.5	8.7	9.6	15.9	3.1	-3.8
Q50	0.5	-5.6	-36.7	7.3	15.4	17.4	20.6	15.3	16.6
Q75	1.1	10.0	-12.1	9.8	19.7	18.4	26.9	27.7	26.5
Q95	1.1	20.3	-93.3	-73.3	8.5	-27.7	5.5	11.8	17.6

Table 6: R_{oos}^2 (%) for Jan 1998–Dec 2024

Model	AR1	UAR	LR	LAS	QRF	QGB	DMQv0c	DMQv1c	DMQv2c
R2	0.5	8.6	-652.5	-6.7	13.3	14.5	7.6	4.3	-1.4

Main Results – Unemployment



Wrapping up

What we did:

1. Develop multi-task neural networks for joint quantile forecasting that respect monotonic quantile ordering
2. Better *out-of-sample* accuracy than quantile ML and econometric models
3. Show quantile forecasts lead to good mean forecasts

What we could do: suggestions?

- ▶ What variables drive the gain in the out-of-sample forecasts
- ▶ Does it hold for other shorter horizon
- ▶ Include non-traditional data (news-based features)
- ▶ Allow for gradual learning of the model parameters after 1997

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Theorem 1 (Quantile Mean Formula) I

Theorem 1. Let $F_t(x) = P(r_{i,t+1} \leq x \mid \mathcal{F}_t)$ be the cumulative distribution function of stock excess return $r_{i,t+1}$ conditional on information set $\mathcal{F}_t$.

For $0 < \tau_1 < \dots < \tau_K < 1$, let

$$\begin{aligned}\bar{Q}_K &= (\tau_1 + \frac{\tau_2 - \tau_1}{2})Q_{\tau_1} + (\frac{\tau_2 - \tau_1}{2} + \frac{\tau_3 - \tau_2}{2})Q_{\tau_2} + \dots \\ &\quad + (\frac{\tau_{K-2} - \tau_{K-1}}{2} + \frac{\tau_{K-1} - \tau_K}{2})Q_{\tau_{K-1}} + (\frac{\tau_K - \tau_{K-1}}{2} + (1 - \tau_K))Q_{\tau_K} \\ &= \frac{\tau_2 + \tau_1}{2}Q_{\tau_1} + \frac{\tau_3 - \tau_1}{2}Q_{\tau_2} + \dots + \frac{\tau_K - \tau_{K-2}}{2}Q_{\tau_{K-1}} + (1 - \frac{\tau_K - \tau_{K-1}}{2})Q_{\tau_K}\end{aligned}$$

where $Q_{\tau_k} = Q_{\tau_k}(r_{i,t+1} \mid \mathcal{F}_t)$ are quantiles of $r_{i,t+1}$, $k = 1, \dots, K$.

Assume that $F_t(x)$ is continuous and strictly monotonically increasing, $E_t(r_{i,t+1})$ exists, and

$$E_t(r_{i,t+1}) = \int_{\mathbb{R}} x d\mu_{F_t},$$

Theorem 1 (Quantile Mean Formula) II

where $\int_{\mathbb{R}} x d\mu_{F_t}$ is a Lebesgue–Stieltjes integral, μ_{F_t} is the Lebesgue–Stieltjes measure associated with F_t , and $\mathbb{R}$ is the set of real numbers. Then

$$\lim_{\Delta_1 \rightarrow 0, \Delta_2 \rightarrow 0} \bar{Q}_K = E_t(r_{i,t+1}),$$

where $\Delta_1 = \max\{\tau_2 + \tau_1, 1 - (\tau_K + \tau_{K-1})\}$ and $\Delta_2 = \max_{2 \leq k \leq K-1} \{Q_{(\tau_k + \tau_{k+1})/2} - Q_{(\tau_{k-1} + \tau_k)/2}\}$.

Proof of Theorem 1 (Sketch)

Proof. Without loss of generality, fix firm i and time t . Since $\int_{\mathbb{R}} x d\mu_{F_t}$ exists, then the Riemann–Stieltjes integral $\int_{\mathbb{R}} x dF_t(x)$ exists and $E_t(r_{i,t+1}) = \int_{\mathbb{R}} x dF_t(x)$. Let $[a, b]$ be an interval containing the support of $r_{i,t+1}$, and partition $[a, b]$ into subintervals. Define the Riemann–Stieltjes sum

$$RS_t(K) = \sum_{k=1}^K c_k (F_t(x_k) - F_t(x_{k-1})),$$

where $c_k \in (x_{k-1}, x_k]$. As the partition gets finer, $RS_t(K) \rightarrow \int_a^b x dF_t(x)$. By choosing quantile points and weights as in the theorem, the sum $\bar{Q}_K$ approximates $E_t(r_{i,t+1})$ arbitrarily well as the mesh of the partition and the quantile spacing go to zero. Thus, $\bar{Q}_K \rightarrow E_t(r_{i,t+1})$ as $K \rightarrow \infty$ and the quantile grid becomes dense.

□